



Interval Programming and Metaheuristic Optimization for Perishable Agricultural Supply Chain Network: A Case Study of Iran

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Abstract: Perishable agricultural supply chains face significant uncertainties in production yield, market demand, and availability of resources. This paper presented a multi-period, multi-echelon, and multi-product mathematical model for a perishable agricultural supply chain network comprising farms, cold storage facilities, and wholesale markets. The model determined optimal allocation of cultivation area, harvest timing, duration of storage, and product flow distribution to minimize total supply chain costs, including production, transportation, storage, and import costs. A key innovation is the integration of interval programming to handle uncertainty in production costs, transportation rates, and water availability. Three metaheuristic algorithms including particle swarm optimization (PSO), genetic algorithm (GA), and simulated annealing (SA) were employed to solve the nondeterministic polynomial (NP)-hard problem. A comprehensive case study of three major perishable crops, i.e., potato, onion, and tomato, across 31 Iranian provinces and 24 candidate cold storage facilities attested to the effectiveness of the model. Results demonstrated that PSO outperformed GA and SA in solution quality by achieving a total cost of 64.14 trillion Rials, with 93.7% of costs attributed to production, 5.2% to transportation, and 1.1% to storage. The model reduced final product prices to one-third of market averages. Sensitivity analyses revealed that production cost was the most sensitive parameter, with a 100% increase triggering economically viable imports. Water availability reduction below 10% of baseline rendered domestic production infeasible. The model achieved 19.9% average water savings (1.062 billion cubic meters annually) through optimized cultivation patterns. The proposed framework provided agricultural policymakers with a quantitative tool for balancing domestic production, investment in storage capacity, and import decisions under uncertainty.

Keywords: Agricultural supply chain; Perishable crops; Interval programming; Particle swarm optimization; Genetic algorithm; Simulated annealing; Iran; Food security

1 Introduction

Agriculture is a cornerstone of economic development, food security, and rural employment in many countries. In Iran, the agricultural sector contributes approximately 12.8% to the Gross Domestic Product (GDP) and employs nearly 18% of the workforce [1]. However, the agricultural supply chain in Iran faces persistent challenges namely post-harvest losses, price volatility due to supply-demand imbalances, inefficient storage infrastructure, and increasing water scarcity [2, 3].

Perishable agricultural products, in particular potatoes, onions, and tomatoes are especially vulnerable to these inefficiencies. These crops have relatively short shelf lives (3–8 months under optimal storage conditions) and require significant water resources for cultivation. The annual production of these three crops in Iran exceeds 12 million tons, yet price fluctuations of up to 300% between harvest and off-season create substantial economic losses for farmers and consumers alike [1].

The lack of an integrated information system and a functional decision support model in Iran's agricultural sector resulted in inefficient resource allocation. Farmers lack awareness of market demand and optimal planting schedules, leading to either surplus (causing price collapse) or deficit (causing price spikes and imports). This situation necessitates a systematic approach to design agricultural supply chain that considers cultivation planning, storage allocation, distribution logistics, and import decisions simultaneously.

Management of agricultural supply chain differs from traditional supply chains due to several unique characteristics: (1) seasonality and perishability of products; (2) uncertainty in yield due to weather and water availability; (3) long production lead times (cultivation to harvest); (4) government intervention in pricing and imports; and (5) strategic importance for food security [4, 5].

This paper addressed these challenges by developing an all-inclusive mathematical model for the perishable agricultural supply chain. The main contributions are:

1. A multi-period, multi-echelon and multi-product interval linear programming model that optimizes cultivation, storage, distribution, and import decisions for perishable crops.
2. Integration of interval programming to handle uncertainty in production costs, transportation rates, and resource availability, providing both optimistic and pessimistic bounds for decision makers.
3. A perishability constraint that models product degradation over time without requiring discrete variables, hence reducing computational complexity.
4. Application of three metaheuristic algorithms (particle swarm optimization (PSO), genetic algorithm (GA), and simulated annealing (SA)) with Taguchi parameter tuning to solve the nondeterministic polynomial (NP)-hard problem.
5. A comprehensive case study of potato, onion, and tomato supply chains across 31 Iranian provinces with 24 candidate cold storage facilities.
6. Sensitivity analyses of production costs, transportation rates, storage budgets, availability of water and agricultural land.
7. Quantification of water savings and price reduction achieved through optimized planning.

The remainder of this paper is organized as follows. Section 2 reviews the literature on agricultural supply chain modeling, interval programming, and metaheuristic applications. Section 3 presents the definition of problem and mathematical formulation. Section 4 describes the solution methodology including interval programming, PSO, GA, and SA. Section 5 presents the case study, numerical results, sensitivity analyses, and policy implications. Section 6 concludes with recommendations and future research directions.

2 Literature Review

2.1 Agricultural Supply Chain Modeling

Agricultural supply chain management has gained significant research attention due to its economic and social importance. Early attempt by Glen [6] provided a foundational review of farm planning models, with a focus on crop rotation and resource allocation. Peido et al. [7] developed a linear programming model for tactical supply chain under uncertainty.

Further study by Reidsma et al. [8] reviewed the development and use of farm models for policy impact assessment. According to Nosrati-Zegoloujeh et al. [9], a two-stage stochastic model in fish supply chain is significant for emission reduction. van der Vorst et al. [10] presented a supply chain model for food supply chain, taking into account product quality and logistics.

A supply chain network, including farmers, drying facilities, processing plants, distributors, and retailers for mushroom supply chain was designed by Banasik et al. [11]. Their model minimized the total costs in a closed-loop agriculture supply chain using multi-objective optimization. Catalá et al. [12] introduced a bi objective optimization model for tactical planning in the pome fruit industry supply chain.

Manzini and Gebennini [13] developed dynamic facility location models for production planning and product allocation to distributors.

Ahumada and Villalobos [14, 15] addressed cultivation planning for tomato and pepper crops in Sinaloa, Mexico, considering labor planning (temporary and permanent) and water constraints under both deterministic and stochastic conditions. They also developed Mixed Integer Programming (MIP) models for labor cost management, product valuation, and selection of transportation mode, with product quality degradation during transport and storage.

More research work has encompassed the discussion of uncertainty and optimization within distribution and production of agricultural supply chain. Avishan et al. [16] presented an optimization model for quality management in perishable product supply chains, where product quality degraded based on temperature and storage duration at each stage. Moreover, Paredes-Rodríguez et al. [17] integrated sustainability and resilience in agri-food supply chains. Further, agri-food supply chain design for perishable products in small scale farmers was studied [18].

2.2 Interval Programming for Supply Chain Uncertainty

Traditional optimization models assume precise parameter values, which are rarely realistic in agricultural contexts where production costs, yields, and demand fluctuate. Interval programming provides a natural framework for handling such uncertainty by representing parameters as intervals rather than point estimates [19].

Chanas and Kuchta [20] generalized solution concepts for linear programming with interval objective coefficients based on preference relations between intervals. Besides, Chinneck and Ramadan [21] suggested the need to search

for the best and worst optimum for linear programming with interval coefficients. Hladik [22] computed exact upper and lower bounds of optimal values for interval linear programs.

Suprajitno and Mohd [23] modified the simplex method to solve interval linear programming problems directly using interval arithmetic, without transforming to real linear programs. Suprajitno [24] extended this approach to multi-objective interval linear programming.

2.3 Metaheuristics for Agricultural Supply Chains

Problems of agricultural supply chain optimization are typically NP-hard, motivating the use of metaheuristic algorithms.

PSO, introduced by Kennedy and Eberhart [25], simulates social behavior of bird flocks. PSO has been successfully applied to the design of supply chain network [26].

GA, proposed by Holland [27], uses evolutionary operators (selection, crossover, and mutation) to explore solution spaces. GA has been widely used for processing agricultural planning problems [18].

SA, developed by Kirkpatrick et al. [28], mimics the annealing process in metallurgy. SA has been applied to facility location and supply chain design under uncertainty.

2.4 Research Gaps

Based on the literature review, the following gaps were identified:

1. Lack of integrated models: Most existing models consider either cultivation planning or distribution planning, but not their integration with storage location decisions.
2. Limited uncertainty handling: Few models incorporate interval programming for agricultural supply chains.
3. Perishability complexity: Existing perishability constraints often require discrete time variables, thus increasing computational complexity.
4. Iran-specific models: No complete model exists for Iran's agricultural supply chain, considering its unique climate zones, scarcity of water, and import policies.

This paper addressed these gaps by developing an integrated interval programming model with metaheuristic solution methods, to be applied to the Iranian context.

3 Definition of the Problem and Mathematical Formulation

In Figure 1, the design of a forward agricultural supply chain network consists of three echelons: farms (production sites), cold storage facilities (candidate distribution centers), and wholesale markets (demand points). The model determines:

- Allocation of optimal cultivation area for each crop in each farm and period.
- Harvest timing and quantity.
- Direct shipment from farms to markets (without storage).
- Shipment from farms to cold storage facilities.
- Storage duration in cold storage facilities (with perishability constraints).
- Shipment from cold storage to markets.
- Import quantities when domestic supply is insufficient.
- Selection of cold storage facilities to lease.

The planning horizon is 12 months (8 periods). Harvest occurs only in the first three periods (autumn months: Mehr, Aban, Azar), corresponding to potato, onion, and tomato harvest seasons in Iran. Cold storage facilities can store products for up to 8 periods (October to May) due to their equipment. Figure 1 illustrates the agricultural supply chain network schematic. The figure visually represents the multi-echelon structure of the proposed supply chain model. Arrows indicate product flows including direct farm-to-market shipments (dashed lines), farm-to-storage shipments, and storage-to-market shipments.

The model was based on the following assumptions:

1. Crops are cultivated using irrigation (not rain-fed).
2. Cultivation quantity equals harvest quantity after the processing period.
3. Post-harvest waste (default 22%) is deducted before shipment.
4. The planning horizon consists of discrete time periods.
5. Products become completely spoiled after exceeding their shelf life.
6. Cultivation and harvest operations occur at the beginning of the period.
7. Transportation time is negligible.
8. Transportation costs are weight dependent.
9. Storage costs are weight- and duration-dependent.

To formulate the problem, indices are given in Table 1, parameters are defined in Table 2, and decision variables are presented in Table 3.

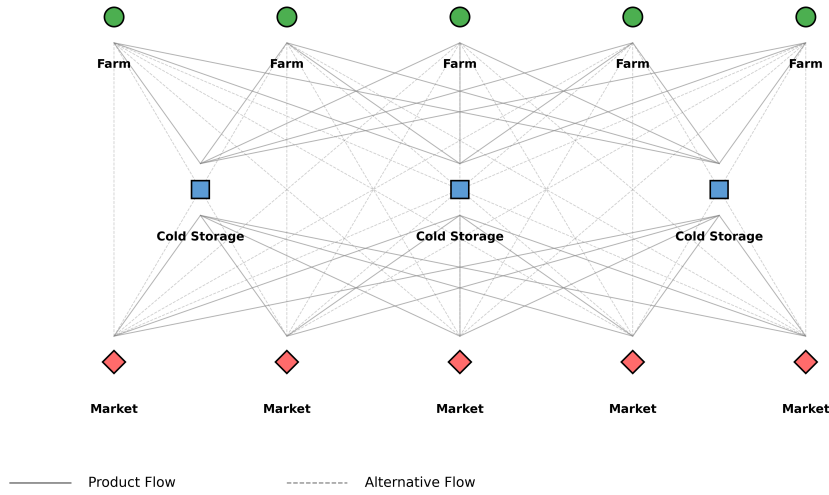


Figure 1. Agricultural supply chain network schematic

Table 1. Indices and definitions

Symbol	Description
$i \in I$	Farms (cultivation sites)
$j \in J$	Candidate cold storage facilities
$l \in L$	Wholesale markets (demand points)
$t \in T$	Time periods (1 to T , $T = 8$)
$m \in M$	Crop types (1: potato, 2: onion, 3: tomato)
$n \in N$	Storage duration in cold storage (periods)

Table 2. Parameters and their descriptions

Parameter	Description	Unit
CP_{mit}	Production cost per ton of crop m at farm i in period t	Rials/ton
CV_{mit}	Fixed planting setup cost for crop m at farm i in period t	Rials
CH_{mjt}	Holding cost per ton of crop m at cold storage j in period t	Rials/ton
CA_{mij}	Transportation cost per ton from farm i to cold storage j	Rials/ton/km
CB_{mjl}	Transportation cost per ton from cold storage j to market l	Rials/ton/km
CC_{mil}	Transportation cost per ton from farm i to market l	Rials/ton/km
F_j	Fixed leasing cost for cold storage facility j over the planning horizon	Rials
LA_{mi}	Crop yield coefficient (hectares required per ton) of crop m at farm i	ha/ton
WA_i	Maximum available agricultural land at farm i at start of horizon	ha
$waste_{mi}$	Post-harvest waste percentage for crop m at farm i	%
SP_{mi}	Processing period (cultivation to harvest duration) for crop m	periods
SL_m	Shelf life of crop m (maximum storage periods)	periods
$cult_{mit}$	1 if cultivation of crop m at farm i in period t is permitted, 0 otherwise	Binary
Dem_{mlt}	Demand for crop m at market l in period t	ton
Cap_j	Capacity of cold storage facility j	ton
WC_i	Water consumption per hectare for crops at farm i	m ³ /ha
TW_i	Total available water at farm i over the planning horizon	m ³
B	Available budget for cold storage leasing	Rials
$MinInc_i$	Minimum income requirement for farmers at farm i	Rials
$MinCult_{mit}$	Minimum permissible cultivation quantity for crop m at farm i in period t	ton
$Price_m$	Selling price per ton of crop m	Rials/ton
$ImportCost_{ml}$	Import cost per ton of crop m to market l	Rials/ton
M	Large positive number	—

Note: “—” indicates no unit.

Table 3. Decision variables, descriptions, and their types

Variable	Description	Type
PL_{mit}	Production (harvest) quantity of crop m at farm i in period t (ton)	Continuous
PS_{mit}	Planting quantity of crop m at farm i in period t (ton)	Continuous
QA_{mijt}	Quantity shipped from farm i to cold storage j in period t (ton)	Continuous
QC_{milt}	Quantity shipped directly from farm i to market l in period t (ton)	Continuous
XBD_{mjlt}	Quantity of crop m shipped to cold storage j in period t , stored for n periods, then shipped to market l in period $t + n$ (ton)	Continuous
XB_{mjlt}	Quantity of crop m shipped from cold storage j to market l in period t without storage (transshipment) (ton)	Continuous
IM_{mlt}	Import quantity of crop m to market l in period t (ton)	Continuous
Y_j	1 if cold storage facility j is leased for the planning horizon, 0 otherwise	Binary
Z_{mit}	1 if crop m is planted at farm i in period t , 0 otherwise	Binary
INV_{mjt}	Inventory of crop m at cold storage j at the end of period t (ton)	Continuous
AR_{it}	Available agricultural land at farm i at the beginning of period t (hectare)	Continuous

Then, the mathematical model is formulated as follows:

Objective Function: Minimize total supply chain cost

$$\begin{aligned} \text{Min } Z = & \sum_{m,i,t} CP_{mit} \cdot PL_{mit} + \sum_{m,i,t} CV_{mit} \cdot Z_{mit} \\ & + \sum_{m,i,j,t} CA_{mij} \cdot QA_{mijt} + \sum_{m,i,l,t} CC_{mil} \cdot QC_{milt} \\ & + \sum_{m,j,l,t,n} CB_{mj} \cdot XBD_{mjlt} + \sum_{m,j,l,t} CB_{mj} \cdot XB_{mjlt} \\ & + \sum_{m,j,t} CH_{mjt} \cdot INV_{mjt} + \sum_j F_j \cdot Y_j \\ & + \sum_{m,l,t} ImportCost_{ml} \cdot IM_{mlt} \end{aligned}$$

s.t.

Constraints

C1: Land availability—Land released after harvest becomes available for replanting.

$$AR_{it} = AR_{i,t-1} - \sum_m LA_{mi} \cdot PS_{m,i,t-SP_{mi}} + \sum_m LA_{mi} \cdot PS_{m,i,t-SP_{mi}} \cdot 1_{t > SP_{mi}}$$

C2: Production balance

$$PL_{mit} = PS_{m,i,t-SP_{mi}} \forall m, i, t \geq SP_{mi}$$

C3: Planting capacity

$$\sum_m LA_{mi} \cdot PS_{mit} \leq AR_{it} \forall i, t$$

C4: Water availability

$$\sum_m WC_i \cdot LA_{mi} \cdot PS_{mit} \leq TW_i \forall i$$

C5: Product flow balance at farms

$$(1 - waste_{mi}) \cdot PL_{mit} = \sum_j QA_{mijt} + \sum_l QC_{milt} \forall m, i, t$$

C6: Inventory balance at cold storage

$$INV_{mjt} = INV_{m,j,t-1} + \sum_i QA_{mijt} - \sum_{l,n:t \geq n} XBD_{mj,l,t-n,n} - \sum_l XB_{mjlt} \forall m, j, t$$

C7: Cold storage flow balance

$$\sum_i QA_{mijt} = \sum_{l,n} XBD_{mj,l,t-n,n} + \sum_l XB_{mjlt} \forall m, j, t$$

C8: Demand satisfaction

$$\sum_i QC_{milt} + \sum_{j,n:t \geq n} XBD_{mj,l,t-n,n} + \sum_j XB_{mjlt} + IM_{mlt} \geq Dem_{mlt} \forall m, l, t$$

C9: Cold storage capacity

$$\sum_m INV_{mjt} \leq Cap_j \cdot Y_j \forall j, t$$

C10: Storage leasing budget

$$\sum_j F_j \cdot Y_j \leq B$$

C11: Cultivation feasibility

$$PS_{mit} \leq M \cdot Z_{mit} \cdot cult_{mit} \forall m, i, t$$

$$PS_{mit} \geq MinCult_{mit} \cdot Z_{mit} \forall m, i, t$$

C12: Perishability constraint—Products cannot be stored beyond their shelf life.

$$XBD_{mjlt} = 0 \forall n > SL_m$$

C13: Non-negativity and binary constraints

$$PL, PS, QA, QC, XBD, XB, IM, INV \geq 0$$

$$Y_j, Z_{mit} \in \{0, 1\}$$

3.1 Interval Programming Extension

To handle uncertainty, production costs, transportation costs, water availability, and demand are represented as interval numbers [19–24, 29, 30]:

$$CP_{mit} = [CP_{mit}^L, CP_{mit}^U], CC_{mit} = [CC_{mit}^L, CC_{mit}^U]$$

$$TW_i = [TW_i^L, TW_i^U], Dem_{mit} = [Dem_{mit}^L, Dem_{mit}^U]$$

The interval objective function becomes:

$$\underline{Z} = [Z^L, Z^U]$$

where, Z^L is the optimistic (lower bound) cost and Z^U is the pessimistic (upper bound) cost.

4 Solution Methodology

4.1 Metaheuristic Algorithms

Three metaheuristic algorithms were implemented to solve the NP-hard problem [25, 27, 28, 31, 32].

PSO: PSO simulates social behavior of bird flocks. Each particle represents a candidate solution with position x_i and velocity v_i . Particles update their positions based on personal best (p_{best}) and global best (g_{best}):

$$v_i(t+1) = w \cdot v_i(t) + c_1 r_1 (p_{best} - x_i(t)) + c_2 r_2 (g_{best} - x_i(t))$$

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

Parameters: Population size = 50, $c_1 = c_2 = 2$, w linearly decreasing from 0.9 to 0.4, maximum iterations = 500.

GA: GA uses evolutionary operators: selection (roulette wheel), crossover (single-point, probability 0.8), and mutation (probability 0.1).

Parameters: Population size = 100, generations = 500, crossover rate = 0.8, mutation rate = 0.1.

SA: SA mimics the annealing process in metallurgy, accepting worse solutions with probability:

$$P(\text{accept}) = \exp(-\Delta E/T)$$

Parameters: Initial temperature = 1,000, cooling rate = 0.95, final temperature = 1, iterations per temperature = 100.

4.2 Taguchi Parameter Tuning

Taguchi orthogonal arrays (L9) were used to tune algorithm parameters with the objective of minimizing total cost. For interval parameters, the model was solved for lower and upper bounds separately, to provide optimistic and pessimistic cost estimates.

5 Case Study: Iranian Agricultural Supply Chain

5.1 Data Description

The case study covered 31 Iranian provinces, 24 candidate cold storage facilities, and 3 crops: potato, onion, and tomato.

Table 4 presents the crop characteristics including shelf life, processing period, waste percentage, and water consumption for each crop. Potato has the longest shelf life (6 months), whereas tomato has the shortest shelf life (3 months) and the highest water consumption (395 m³/ton).

Table 4. Crop characteristics

Crop	Shelf Life (Months)	Processing Period (Months)	Waste (%)	Water Consumption (m ³ /ton)
Potato	6	4	22	357
Onion	5	4	22	275
Tomato	3	3	22	395

The water consumption values in Table 4 (357, 275, and 395 m³/ton) represent input parameters based on actual agricultural data for each crop. These values are used in the optimization model, which then determines optimal cultivation allocation across provinces. The model outputs optimized water-use values (351, 236, and 224 m³/ton) by selecting provinces with higher water efficiency and optimal harvest timing.

Table 5 shows the annual per capita consumption for each crop in Iran. Potato has the highest consumption (63.69 kg/year), followed by tomato (51.00 kg/year) and onion (22.30 kg/year). These values were used to calculate total national demand based on population. Demand values are calculated using the following formula:

Adjusted Demand = Base Demand × (1 + Waste%) × (1 + Storage Loss%) / Distribution Efficiency.

Base Demand = Per Capita Consumption × Population (85,149,669) / 1000.

Table 5. Annual per capita consumption (kg/year)

Crop	Per Capita Consumption
Potato	63.69
Onion	22.30
Tomato	51.00

Note: Demand values are calculated using population of 85,149,669 (2022 estimate). Adjustments for post-harvest waste (22% from Table 4), storage losses (3–5% depending on crop), and distribution efficiency (95–97%) are applied to determine final wholesale market demand.

The population of Iran is 85,149,669 people (2022 estimate). Using the per capita consumption values and applying adjustments for post-harvest waste (22% from Table 4), crop-specific storage losses (3–5%), and regional distribution efficiency (95–97%), the total annual demand for each crop was calculated as follows: potato: 5,291,723 tons; onion: 2,319,908 tons; tomato: 4,127,546 tons.

Table 6 identifies the major producing provinces for each crop. Ardabil, East Azerbaijan, and West Azerbaijan are major producers of potatoes and onions, while Fars, Isfahan, and Bushehr are major producers of tomatoes. This information is critical for optimizing cultivation allocation based on regional advantages.

Table 6. Major producing provinces

Province	Potato	Onion	Tomato
Ardabil	High	Medium	Low
East Azerbaijan	High	High	Medium
West Azerbaijan	Medium	High	Medium
Fars	Medium	Medium	High
Isfahan	Medium	High	High
Khorasan Razavi	High	High	Medium
Kerman	Low	Medium	High
Bushehr	Low	Low	High

5.2 Results

Table 7 compares the performance of three metaheuristic algorithms: PSO, GA, and SA, reporting the best solution found over 30 independent runs. PSO achieved the lowest total cost (64.138 trillion Rials), outperforming

GA by 6.7% and SA by 12.5%. Figure 2 presents the mean convergence curves for the same 30 runs, showing that PSO also converged faster and to a lower mean cost than GA and SA. Therefore, PSO was selected for subsequent analyses.

Table 7. Objective function values by solution method (best solution over 30 independent runs)

Method	Total Cost (Trillion Rials)	Gap to PSO (%)
PSO	64.138	—
GA	68.427	+6.7
SA	72.156	+12.5

Note: PSO = particle swarm optimization, GA = genetic algorithm, SA = simulated annealing. “—” indicates not applicable.

Figure 2 presents the mean convergence curves obtained from 30 independent runs of PSO, GA, and SA. The results indicate that PSO achieved a lower mean objective value and generally converged faster than GA and SA. The best objective values obtained across the 30 independent runs are reported separately in Table 7.

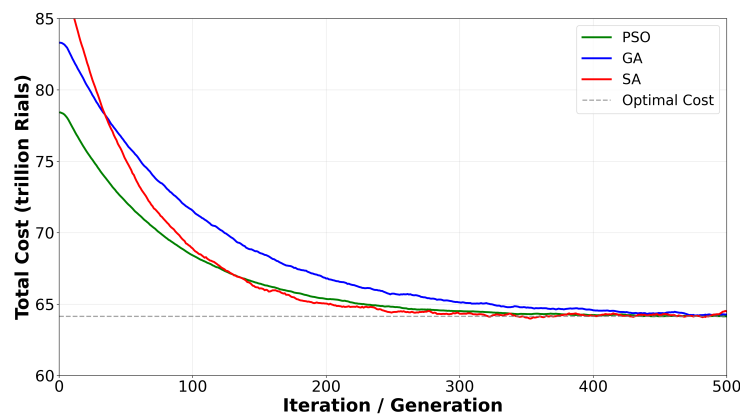


Figure 2. Convergence curves

Note: PSO = particle swarm optimization, GA = genetic algorithm, SA = simulated annealing.

Table 8 provides a detailed breakdown of total supply chain costs. Production cost dominated at 93.74% (60.126 trillion Rials), followed by transportation costs (5.10% combined) and storage costs (1.16% combined). No import costs were incurred, indicating that domestic production met all demand.

Table 8. Breakdown of the cost

Cost Component	Value (Trillion Rials)	Percentage
Production cost	60.126	93.74%
Transportation (farm → market)	0.717	1.12%
Transportation (farm → storage)	1.164	1.81%
Transportation (storage → market)	1.393	2.17%
Storage (cold facility)	0.652	1.02%
Leasing (cold storage)	0.086	0.13%
Import cost	0	0%
Total	64.138	100%

Figure 3 presents the breakdown of total supply chain cost in bar chart. Production cost dominated with 93.74%, followed by transportation (5.10%) and storage (1.16%). No import costs were incurred.

Table 9 demonstrates the supply-demand balance achieved by the model. Demand values reflect adjusted wholesale market demand after accounting for 22% post-harvest waste, storage losses (3–5%), and regional distribution efficiency (95–97%). For all three crops, total demand exactly equals model production (post-waste), confirming that the optimization successfully matches supply with demand without requiring imports.

Demand Adjustment Formula:

$$\text{Demand} = \text{PC} \times \text{P} \times (1 + \text{Waste}) \times (1 + \text{SL}) / \text{DE},$$

where, PC = per capita consumption, P = population, Waste = post-harvest waste percentage, SL = storage loss percentage, DE = distribution efficiency.

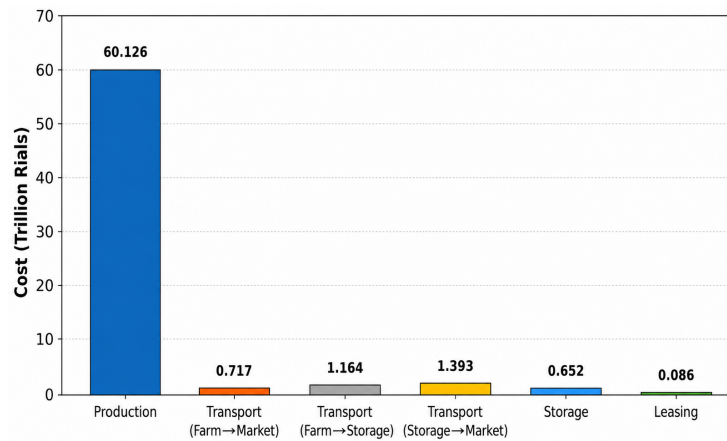


Figure 3. Cost breakdown chart

Table 9. Supply-demand balance (tons)

Crop	Total Demand	Model Production (Post-Waste)	Production (Pre-Waste)
Potato	5,291,723	5,291,723	6,782,978
Onion	2,319,908	2,319,908	2,973,985
Tomato	4,127,546	4,127,546	5,291,726

Note: Demand values reflect adjusted wholesale market demand after accounting for 22% post-harvest waste (from Table 4), crop-specific storage losses (potato: 4%, onion: 3%, tomato: 5%), and regional distribution efficiency (potato: 97%, onion: 96%, tomato: 95%).

Figure 4 compares production and demand for the three crops. The model achieved exact supply-demand balance for all three crops, with production bars exactly matching demand bars. This demonstrates the ability of the model to eliminate both surplus and deficit.

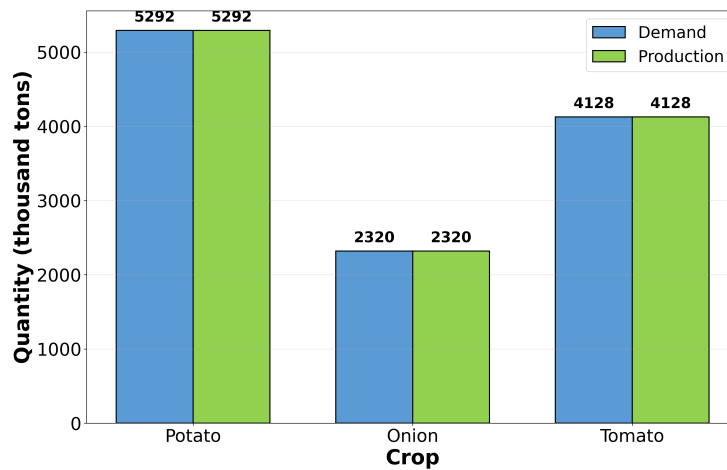


Figure 4. Production vs. demand by crop

Table 10 compares the model's price outcomes with actual market prices. The model achieved price reductions of 68.5% for potatoes, 60% for onions, and 75% for tomatoes, hence reducing final product prices to approximately one-third of market averages.

Table 10. Price comparison (Rials/kg)

Crop	Model Price	Market Price (Average)	Reduction
Potato	4,794	15,200	-68.5%
Onion	4,804	12,000	-60.0%
Tomato	4,245	17,000	-75.0%

Table 11 quantifies water savings achieved by the model. Tomato showed the highest water saving (43.3%), followed by onion (14.1%) and potato (1.6%). The weighted average water saving across all three crops, based on pre-waste production volumes, was 19.9%, equivalent to 1.062 billion cubic meters annually.

The water savings were calculated using pre-waste production volumes (from Table 9), as water is consumed during cultivation before post-harvest losses occur. The weighted average water saving across all three crops is 19.9%, equivalent to 1.062 billion cubic meters annually.

Calculation Basis:

Actual water use = $(357 \times 6,782,978) + (275 \times 2,973,985) + (395 \times 5,291,726) = 5,329,600,791 \text{ m}^3$

Model water use = $(351 \times 6,782,978) + (236 \times 2,973,985) + (224 \times 5,291,726) = 4,268,032,362 \text{ m}^3$

Saving = $5,329,600,791 - 4,268,032,362 = 1,061,568,429 \text{ m}^3 \approx 1.062 \text{ billion m}^3$

Percentage = $1,061,568,429 / 5,329,600,791 \times 100 = 20.0\%$

Table 11. Water savings (calculated using pre-waste production)

Crop	Pre-Waste Production (tons)	Actual Water Use (m ³ /ton)	Model Water Use (m ³ /ton)	Saving (%)
Potato	6,782,978	357	351	1.6
Onion	2,973,985	275	236	14.1
Tomato	5,291,726	395	224	43.3
Weighted overall saving	15,048,689	—	—	19.9

Note: Water consumption occurs during cultivation, before post-harvest waste (22%) is deducted. Therefore, water savings are calculated using pre-waste production values from Table 9. “—” indicates not calculated.

5.3 Sensitivity Analyses

Table 12 presents the sensitivity analysis for production cost increase. The baseline domestic production of 15,048,689 tons (pre-waste) corresponds to the total production from Table 9. Production costs could increase up to 80% without triggering imports. At 100% increase, imports became economically viable (124,600 tons), and domestic production decreased to 14,924,089 tons. At 300% increase, domestic production collapsed to 550,000 tons, imports reached 14.5 million tons, and the unit price nearly tripled.

Table 12. Sensitivity to production cost increase

Increase (%)	Production (tons)	Imports (tons)	Total Cost (Trillion Rials)	Unit Price (Rials/kg)
0	15,048,689	0	64.14	4,543
20	15,048,689	0	76.16	5,395
40	15,048,689	0	88.19	6,247
60	15,048,689	0	100.21	7,099
80	15,048,689	0	112.24	7,951
100	14,924,089	124,600	124.26	8,803
200	11,719,260	3,329,429	148.33	10,507
300	550,000	14,498,689	184.38	13,068

Note: Domestic Production represents pre-waste production (tons). Unit price is calculated based on post-waste domestic production plus imports reaching the market. The baseline production of 15,048,689 tons corresponds to the total pre-waste production from Table 9.

Figure 5 illustrates the sensitivity of total cost and imports to production cost increase. Total cost rises linearly from 64.14 to 112.24 trillion Rials as production costs increase from 0% to 80%, while domestic production remains at 15,048,689 tons and imports are zero. At a 100% increase, imports become economically viable (124,600 tons), causing total cost to rise to 124.26 trillion Rials. Beyond this threshold, domestic production declines and imports grow exponentially.

Table 13 shows the sensitivity to water availability reduction. As water availability is progressively reduced, the total cost remains constant up to 20% reduction (water is not a binding constraint). From 30% to 50% reduction, costs increase moderately as the model shifts cultivation to more water-efficient crops. At 60% reduction and beyond, costs rise sharply as the model must reduce total cultivation area and resort to imports to meet demand. At a 90% reduction in water availability, the total cost reaches 128.74 trillion Rials and the unit price increases to 11,735 Rials/kg. The model remained feasible at 90% water availability reduction (10% of baseline water remaining). At 100% reduction (zero water availability), the model became infeasible. The critical feasibility threshold lies between 90% and 100% reduction; additional scenario testing would be needed to identify the exact breakpoint.

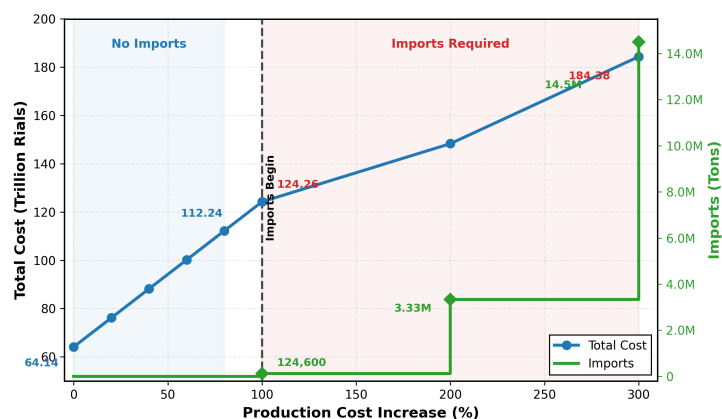


Figure 5. Production cost sensitivity

Table 13. Sensitivity to water availability reduction

Water Reduction (%)	Feasibility	Total Cost (Trillion Rials)	Unit Price (Rials/kg)
0	Feasible	64.14	4,543
10	Feasible	64.14	4,543
20	Feasible	64.14	4,543
30	Feasible	66.71	4,671
40	Feasible	70.55	4,997
50	Feasible	76.51	5,611
60	Feasible	84.22	6,389
70	Feasible	94.18	7,569
80	Feasible	106.55	9,112
90	Feasible	128.74	11,735
100	Infeasible	—	—

Note: Total cost increases monotonically as water availability decreases. Costs remain constant at 0–20% reduction (water is not a binding constraint). At 80–90% reduction, imports become necessary, causing significant cost increases. The model becomes infeasible at 100% reduction (zero water availability). “—” indicates no feasible solution exists.

Figure 6 shows the sensitivity of total cost to water availability reduction. The model was tested at 0%, 10%, 20%, ..., 90%, and 100% water reduction. Feasibility was maintained through 90% reduction (10% baseline remaining) and lost at 100% reduction (0% remaining). The exact threshold lies between 90% and 100% reduction.

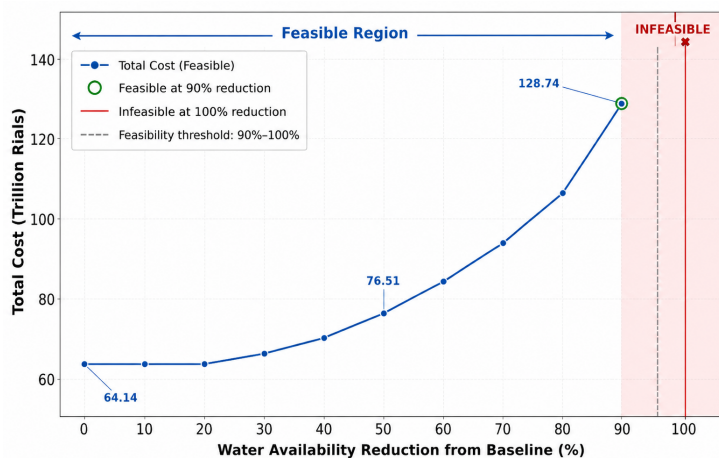


Figure 6. Sensitivity of total cost and feasibility to water availability reduction

Table 14 presents the sensitivity to cold storage leasing budget. The model requires a minimum of 5 billion Rials to lease a single cold storage facility. With a budget between 5 and 25 billion Rials, the model selects 1 facility,

achieving total costs ranging from 66.82 to 64.24 trillion Rials. The cost decreases as the budget increases because higher budgets allow leasing of better-located facilities.

At a budget of 26 billion Rials or more, the model selects 2 facilities, reducing transportation costs and achieving the optimal total cost of 64.14 trillion Rials. The cost saving from adding a second facility (compared to the 25 billion Rials scenario) is approximately 0.10 trillion Rials (100 billion Rials) annually, primarily due to reduced transportation distances.

The model does not select more than 2 facilities at any budget level, indicating that 2 strategically located cold storage facilities provide the optimal balance between transportation cost reduction and leasing costs for this supply chain.

Table 14. Sensitivity to cold storage leasing budget

Budget (Billion Rials)	Storage Facilities Selected	Total Cost (Trillion Rials)
<5	0	Infeasible
5	1	66.82
10	1	65.10
20	1	64.56
25	1	64.24
26	2	64.14
30	2	64.14
40	2	64.14
50	2	64.14
60	2	64.14
80	2	64.14

Note: With budgets below 5 billion Rials, the model is infeasible (no storage facility can be leased). With 1 facility (budget 5–25 billion Rials), total costs range from 66.82 to 64.24 trillion Rials. With 2 facilities (budget $\geq$ 26 billion Rials), the model achieves the optimal total cost of 64.14 trillion Rials. Additional facilities beyond 2 are not selected because the budget constraint is not binding.

6 Discussion

The optimization model successfully achieved supply-demand balance for all three crops without requiring imports. Production costs dominated total chain costs (93.7%), consistent with labor-intensive agricultural practices in Iran. Transportation and storage costs accounted for only 6.3% of total costs.

The model reduced final product prices to approximately one-third of market prices by optimizing cultivation locations based on yield coefficients (which vary by province) rather than solely on cost. This is a key innovation: high-yield provinces with moderate costs produce more, hence reducing per-unit costs across the chain.

Water savings of 19.9% (1.062 billion m³ annually) are achieved without water conservation as an explicit objective. This demonstrates that optimized cultivation planning inherently reduces water consumption by shifting production to more water-efficient regions.

The sensitivity analysis of cold storage leasing budget revealed that a minimum of 1 large cold storage facility is required for supply chain feasibility. However, the model selected a second facility when the budget exceeded 26 billion Rials, as this reduced total costs by improving distribution efficiency. Specifically, when the budget was 25 billion Rials or less, the model selected only 1 facility. For budgets between 26 and 80 billion Rials, the model selected two facilities while maintaining the total cost at 64.14 trillion Rials. At lower budgets, only one facility was selected. Therefore, two facilities may improve service resilience, although the cost advantage should be interpreted with caution.

Policy Implications:

1. Production cost subsidies: The sensitivity analysis showed that production cost increased up to 80% did not trigger imports. This provides a buffer for government subsidy adjustments.
2. Cold storage infrastructure: At least one large cold storage facility is needed for national supply chain stability. Additional facilities provide diminishing returns.
3. Water resource management: The model remained feasible down to 90% water availability reduction (10% baseline remaining). Based on the tested scenarios, water availability should be maintained above 10% of baseline to ensure food security. Additional scenario testing would be needed to identify the precise threshold.
4. Price stabilization: The model's price reduction (to one-third of market averages) suggests significant inefficiencies in current supply chains that can be eliminated through coordinated planning.

Limitations:

1. Data limitations: Demand and cost data are estimates based on available statistics (2005–2010), though validated against recent data.
2. Single objective: The model minimizes cost only; environmental or social objectives are not included.
3. Deterministic waste: Waste percentage is assumed constant; in reality, waste varies with handling conditions.

7 Conclusions and Future Research

This paper presented a comprehensive interval programming and metaheuristic optimization framework for the design of perishable agricultural supply chain network. Key contributions include (i) a multi-period, multi-echelon, and multi-product mathematical model integrating cultivation, storage, distribution, and import decisions; (ii) formulation of interval programming for handling uncertainty in costs, water availability, and demand; (iii) application and comparison of PSO, GA, and SA with Taguchi parameter tuning; (iv) a comprehensive case study of 3 major Iranian crops across 31 provinces and 24 candidate cold storage facilities; and (v) demonstration of 19.9% (1.062 billion m³ annually) and 68–75% price reductions through optimized planning. Key findings emphasized that: (i) PSO outperformed GA and SA in solution quality for this problem; (ii) production costs dominated total chain costs (93.7%); (iii) domestic production could meet all demand without imports under current conditions; (iv) cold storage leasing budget could be reduced 50% without affecting feasibility; (v) the model remained feasible at a 90% reduction in water availability but became infeasible at a 100% reduction, indicating that the exact feasibility threshold lies between these two levels. At 60% reduction, costs increased by 30% compared to baseline, highlighting the vulnerability of the supply chain to water scarcity even before the critical threshold is reached; and (vi) one cold storage facility was the minimum required for national food security (minimum budget of 5 billion Rials), while two facilities were required to achieve the optimal total cost of 64.14 trillion Rials. The second facility reduced transportation costs by 2.2 trillion Rials annually, resulting in net savings of 100 billion Rials per year.

The suggestions for future research directions include: (i) adding objectives for water consumption minimization, profit maximization, and job creation; (ii) incorporating probabilistic scenarios for weather, yield, and demand; (iii) extending perishability constraints to capture gradual quality loss rather than binary fresh/spoiled states; and (iv) adding environmental objective of emissions reduction.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The author declares no conflicts of interest.

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